Control, Filtering, and Interest-Rate Smoothing: A Note on Policy under Transmission Uncertainty

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Abstract

This note illustrates the duality between linear-quadratic control and Kalman filtering in a simple monetary-policy setting. Reformulating a standard policy optimization problem as a filtering problem provides an alternative interpretation of interest-rate smoothing. Increased uncertainty regarding the effectiveness of policy actions is shown to be mathematically equivalent to placing an explicit penalty on variability in the policy instrument. Consequently, interest-rate smoothing need not reflect an intrinsic preference for stable policy rates. Instead, it can be interpreted as a rational response to uncertainty surrounding the monetary-policy transmission mechanism.

1 Introduction

This note illustrates how a control problem can be reformulated as a filtering problem. Such a reformulation makes control problems amenable to estimation through the Kalman filter in a relatively straightforward manner, although that aspect is not pursued further here. Instead, the focus of this note is on a different insight. Expressing a control problem as a filtering problem makes it transparent that a reduction in the signal-to-noise ratio, which in the filtering framework corresponds to increased uncertainty regarding the effects of stabilization policy, is mathematically equivalent to introducing a penalty on variability in the control variable in an augmented objective function of a standard control problem.

2 A Simple Control Problem

We begin by considering a simple model in which a central bank seeks to minimize deviations of inflation from its target,

$$(\pi_t - \pi^*) = \hat{\pi}_t,$$

as well as deviations of output from its long-run sustainable level,

$$y_t - y^* = x_t.$$

The central bank is assumed to minimize the following loss function:

$$\sum_{t=0}^{\infty} 0.5\, \beta^t E_{t|t_0} \left(\hat{\pi}_t^2 + \lambda x_t^2 \right), \qquad \lambda = 1, \quad \beta \approx 1. \tag{1}$$

The relationship between inflation and the output gap is described by the following simple model:

$$\hat{\pi}_{t+1} = 0.75\hat{\pi}_t + 0.20x_t + u_{t+1}^S, \tag{2}$$

$$x_{t+1} = 0.25x_t - 0.25(\hat{\imath}_t - \hat{\pi}_t) + u_{t+1}^D, \tag{3}$$

where u^S_{t+1} and u^D_{t+1} denote supply and demand shocks, respectively. Equations (2) and (3) may also be written in compact form as

$$\underbrace{\begin{pmatrix} \hat{\pi}_{t+1} \\ x_{t+1} \end{pmatrix}}_{w_{t+1}} = \underbrace{\begin{pmatrix} 0.75 & 0.20 \\ 0.25 & 0.25 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} \hat{\pi}_t \\ x_t \end{pmatrix}}_{w_t} + \underbrace{\begin{pmatrix} 0 \\ -0.25 \end{pmatrix}}_{B} \hat{i}_t + \underbrace{\begin{pmatrix} u^S_{t+1} \\ u^D_{t+1} \end{pmatrix}}_{\varepsilon_{t+1}}. \tag{4}$$

The solution to the control problem yields the following interest-rate response rule:

$$\hat{i}_t = 1.86\,\hat{\pi}_t + 1.23\,x_t. \tag{5}$$

3 Duality Between Control and Filtering

It is well known that control and filtering are dual problems.[1] This implies that a control problem can be reformulated as a filtering problem through the Kalman filter. A state-space system can be written in the general form:[2]

$$a_{t+1} = Ta_t + R\varepsilon^a_{t+1}, \qquad \mathrm{Var}(\varepsilon^a_t) = Q, \tag{6}$$

$$y_t = Za_t + H\varepsilon^y_t, \qquad \mathrm{Var}(\varepsilon^y_t) = H, \tag{7}$$

In the Kalman filter, the signal-to-noise ratio refers to the relative size of the state-innovation variance Q to the measurement-error variance H; a higher ratio implies that observations are informative, while a lower ratio implies that measurement noise dominates. This ratio determines the magnitude of the Kalman gain and therefore how strongly new observations affect the filtered estimate. When

$$T = A' = \begin{pmatrix} 0.75 & 0.25 \\ 0.20 & 0.25 \end{pmatrix}, \qquad Z = B' = \begin{pmatrix} 0 & -0.25 \end{pmatrix},$$

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix}, \qquad H = 0,$$

the Kalman gain becomes

$$\hat{i}_t = 1.86\,\hat{\pi}_t + 1.23\,x_t = Fw_t. \tag{8}$$

That is, the Kalman filter produces the same result as equation (5), illustrating the duality between control and filtering.

4 Control, Filtering, and Uncertainty about the Response Rule

What happens if a disturbance term is introduced into the optimization problem underlying equation (8)? One simple interpretation is that economic activity is affected by actual market interest rates rather than directly by the central bank's policy rate, and that the central bank cannot perfectly influence the market rate. In that case, equation (5) is modified to

$$\underbrace{\hat{i}^m_t}_{\text{actual interest rate}} = F \underbrace{w_t}_{\text{policy rate}} + \underbrace{\varepsilon^i_t}_{\text{disturbance}}, \tag{9}$$

[1] That is, the Kalman filter is dual to a control problem with a quadratic loss function and a linear constraint. In this note, the filtering problem is assumed to be time-invariant, which corresponds directly to an infinite-horizon control problem. However, there is no reason why this must necessarily be the case when estimating a finite-horizon control problem. In such a setting, the standard Kalman filter is likely to remain the dual representation, although this remains to be investigated.

[2] See Andrew Harvey, *Forecasting, Structural Time Series Models and the Kalman Filter*, Cambridge University Press, 1989, for a discussion of Kalman filtering and state-space systems.

where

$$\varepsilon_t^i \sim N_{iid}(0, 0.25).$$

In the filtering problem, this case can be represented simply by setting

$$H = 0.25.$$

For the parameter values considered here, the Kalman filter then yields the solution

$$\hat{i}_t = 0.55\hat{\pi}_t + 0.32x_t. \tag{10}$$

Exactly the same solution is obtained if the objective function in the standard control problem is extended to account for variability in the control variable. In our example, the loss function in equation (1) is replaced by

$$\sum_{t=0}^{\infty} 0.5\, \beta^t E_{t|t_0} \left(\hat{\pi}_t^2 + \lambda x_t^2 + \mu \hat{i}_t^2\right), \qquad \lambda = 1, \quad \mu = 0.25. \tag{1'}$$

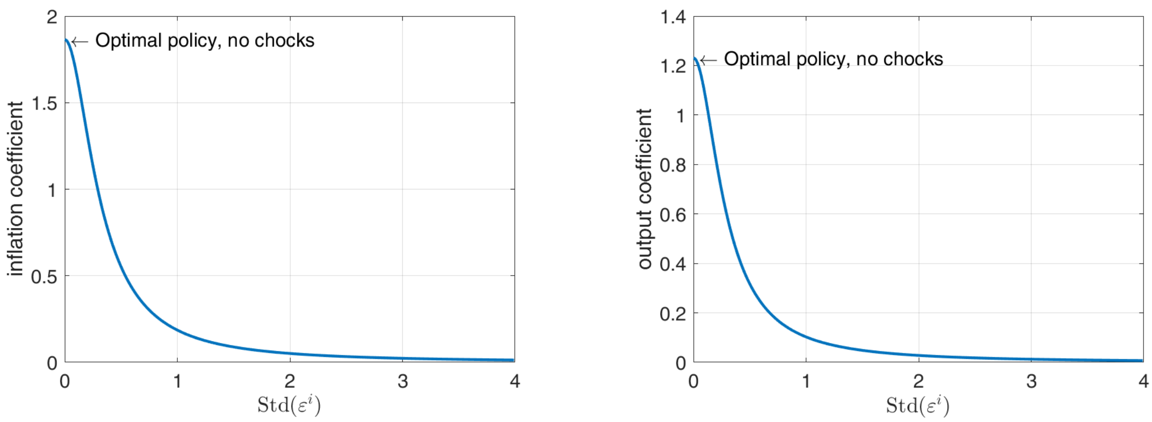
Minimizing (1') yields an interest-rate reaction function that is identical to the Kalman filter solution once account is taken of the fact that

$$\varepsilon_t^i \sim N_{iid}(0, 0.25),$$

namely

$$\hat{i}_t = 0.55\hat{\pi}_t + 0.32x_t. \tag{10'}$$

As $H \to 0$, uncertainty regarding the transmission mechanism vanishes and the signal-to-noise ratio becomes arbitrarily large. The Kalman gain therefore converges to the benchmark optimal-policy solution. As variability in the control variable increases relative to the variability of the target variables, less weight is placed on deviations of the target variables in the reaction function. Put differently, greater uncertainty regarding the effectiveness of policy actions leads to a more cautious policy response. This relationship is illustrated in the figure below.



(a) Inflation coefficient as a function of transmission uncertainty.

(b) Output coefficient as a function of transmission uncertainty.

Figure 1: Increasing transmission uncertainty reduces the responsiveness of the optimal policy rule. In the dual control formulation, the same effect is obtained by increasing the weight placed on interest-rate variability.

5 Conclusion

This note has illustrated the duality between a simple monetary-policy control problem and its corresponding filtering representation. Introducing uncertainty regarding the transmission of monetary policy reduces the Kalman gain and leads to a more cautious policy response. Exactly the same outcome can be obtained in the control problem by placing a positive weight on variability in the policy instrument. The result suggests that interest-rate smoothing and transmission uncertainty may be viewed as alternative representations of the same underlying optimization problem.